

Scope and Instructions

This version is conceptual-first and covers electric forces and fields, oscillations, waves, and sound. Explain direction, limiting behavior, symmetry, and physical meaning whenever asked. Use the word bank only for Questions 1 and 2; some words appear more than once and some do not belong.

Suggested constants and reminders:
 $k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$ $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/(\text{N}\cdot\text{m}^2)$
 $v = f\lambda$ $\omega = 2\pi f$ $T = 1/f$ $F = -kx$ $\oint \mathbf{E} \cdot d\mathbf{A} = Q_{\text{enc}}/\epsilon_0$

Answer in complete physics statements. A correct diagram must include axes, directions, and labels without relying on color alone.

1. Electric-field word bank

Fill each blank using the word bank. Terms may be used more than once; several are distractors. Cross out each term as you use it.

WORD BANK

source test enclosed external scalar vector
tangent perpendicular inverse-square conserved canceled
reinforced zero nonzero frequency wavelength

- a. The electric field is force per unit _____ charge.
- b. Field direction is defined using a positive _____ charge.
- c. The net electric field from several charges is a _____ sum.
- d. In a conductor at electrostatic equilibrium, E inside the metal is _____.
- e. Gauss flux depends on _____ charge.
- f. Field lines meet a conductor surface _____.
- g. Coulomb magnitude has an _____ dependence on distance.
- h. Charge is _____ in an isolated system.
- i. At a midpoint, symmetric components may be _____ or _____.
- j. A field-line arrow tells the force direction for a _____ test charge.
- k. If the system is not isolated, charge can cross the boundary but the enlarged system still has _____ charge.

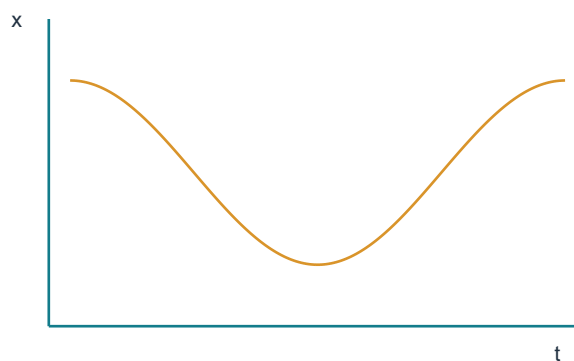
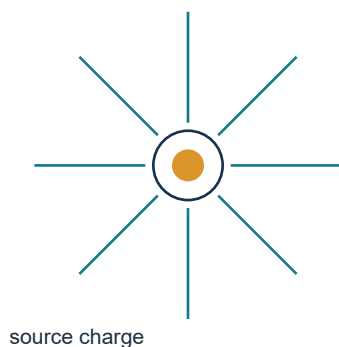
Briefly explain one answer: identify the physical principle, not just the vocabulary word.

2. Matching: visual behavior to equation

Match each visual/phenomenon (A-F) to one equation (1-6). Use each equation once.

A. Field lines radiate from an isolated positive charge. B. A closed surface counts net field-line excess. C. A mass oscillates about equilibrium. D. A shape translates right unchanged. E. A potential well has a horizontal minimum. F. A sound level compares a quantity to a reference logarithmically.

1. $F = -dU/dr$ 2. $\beta = 10 \log_{10}(I/I_0)$ 3. $E = kQ/r^2$ 4. $\oint \mathbf{E} \cdot d\mathbf{A} = Q_{\text{enc}}/\epsilon_0$ 5. $y = f(x-vt)$ 6. $x'' + \omega^2 x = 0$



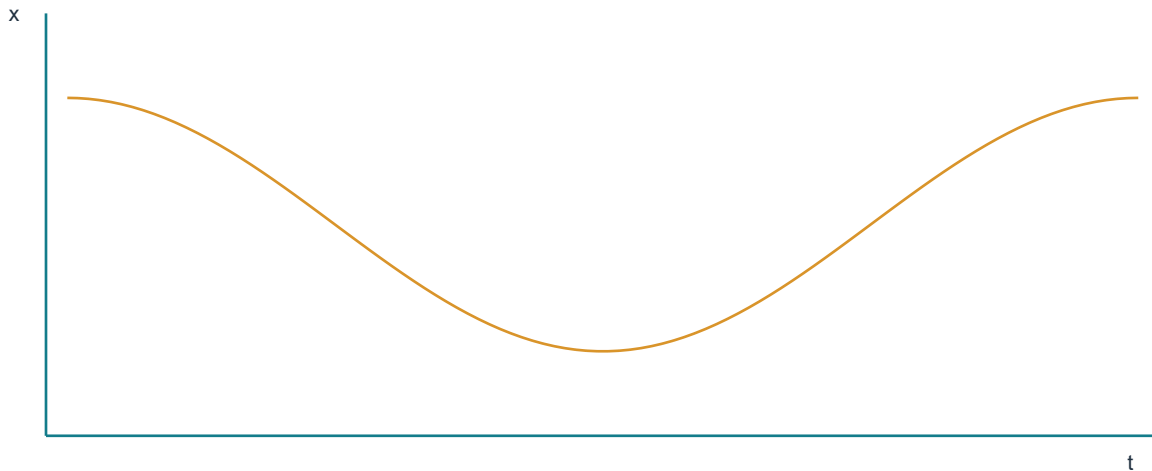
3. Explain electric-field phenomena

Answer in words and include a small labeled sketch for each.

- a. Why does a negative test charge accelerate opposite the direction of E ?
- b. Why can two nonzero electric fields produce zero net field at a point?
- c. Why is the field inside conducting material zero at electrostatic equilibrium, and why is the surface field normal?
- d. Explain why an external charge can change field values on a Gaussian surface without changing the net flux through it.
- e. What is the difference between a field existing and a test charge experiencing a force? f. Why is E measured in N/C rather than newtons alone?

4. Draw and interpret SHM graphs

For $x(t) = A \cos(\omega t)$, draw on shared axes one cycle of $x(t)$, $v(t)$, and $a(t)$, with relative phase and amplitudes labeled. Then draw $U_s(x)$ and $K(x)$ for an ideal mass-spring oscillator. Also identify the sign of a and the direction of the restoring force at $x < 0$. Explain why the period does not depend on amplitude in ideal SHM. Identify where $x=0$, $x=\pm A$, v is maximum, and acceleration is maximum in magnitude.

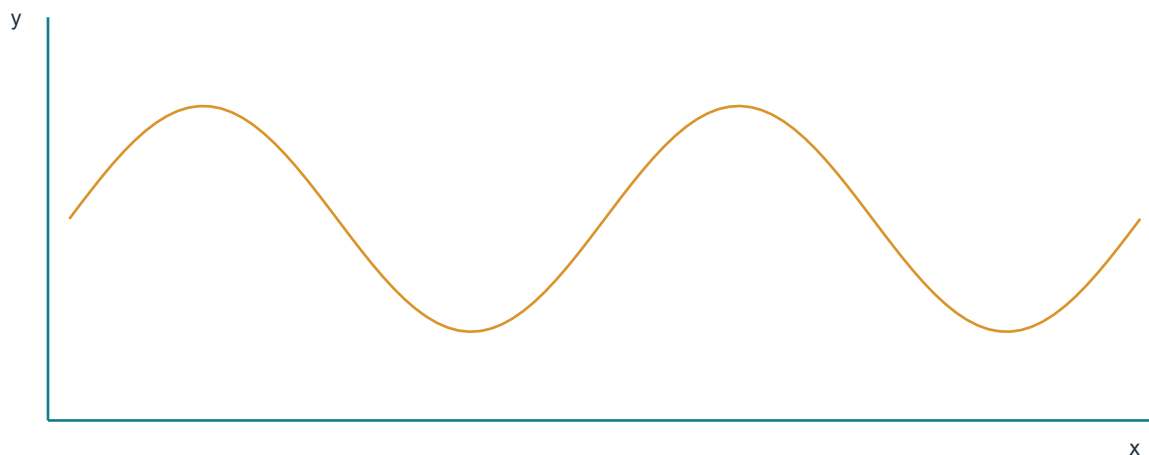


5. Identify the best wave graph

A traveling-wave snapshot is described by $y=A \cos(kx-\omega t)$. Choose which behavior is physically correct and explain why:

- I. Increasing A changes wave speed in every ideal linear medium.
- II. Increasing k decreases wavelength.
- III. Reversing the sign before ωt changes the propagation direction.
- IV. The medium's particles travel with the same speed as the wave pattern.

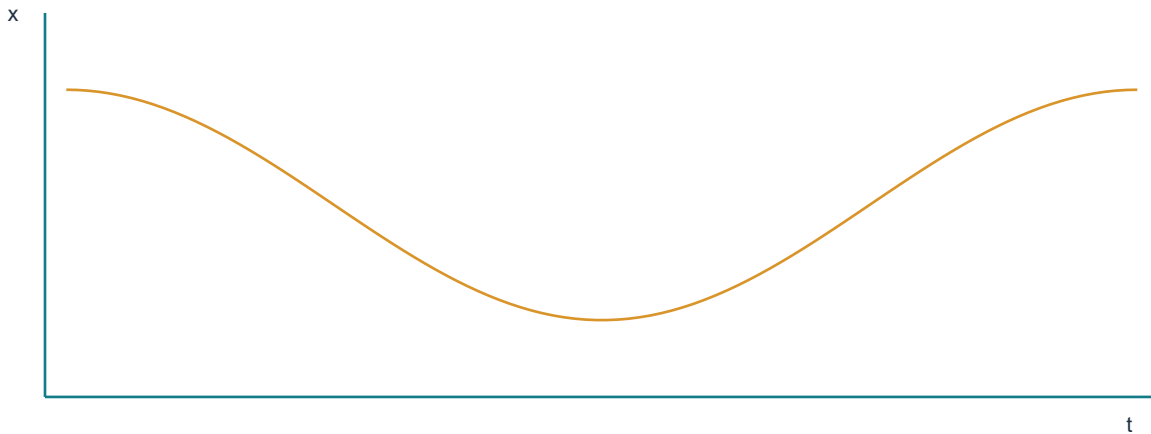
Select all correct statements and draw a labeled snapshot showing λ and A . Then state what a fixed point on the medium does as the wave passes, and explain why a crest is not a material object moving through the medium.



6. Oscillations: conceptual comparisons

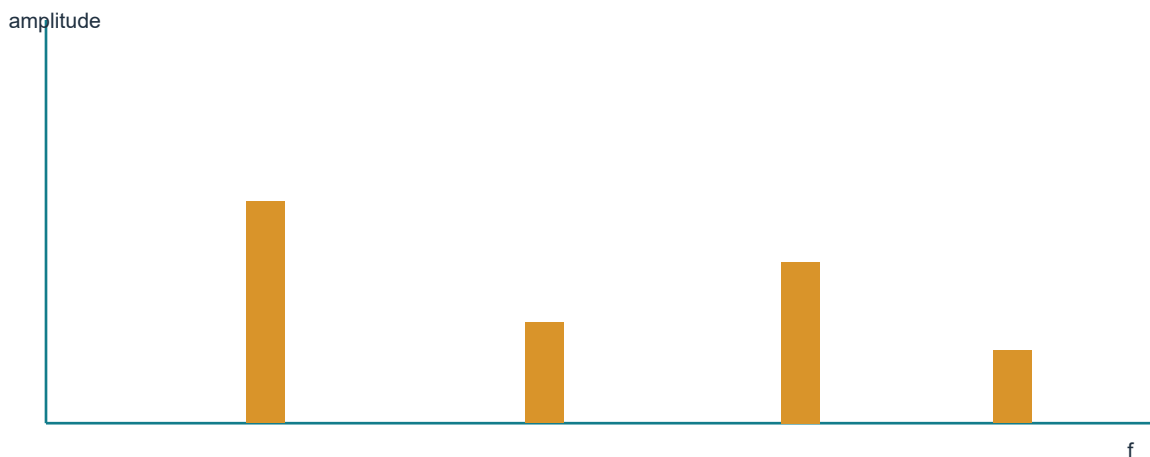
For each pair, state which is larger or whether they are equal, and justify.

- Two identical springs: one has twice the mass. Compare periods.
- Same oscillator at $x=0$ and $x=A$. Compare speed and spring potential energy.
- Two oscillators have the same amplitude but different k . Compare total energy.
- A damped oscillator is driven at frequencies below, near, and above resonance. Describe the qualitative amplitude trend. Define resonance in terms of energy transfer, explain what damping changes, and state whether the natural frequency is set by the driving frequency or by the system.



7. Sound and spectrum

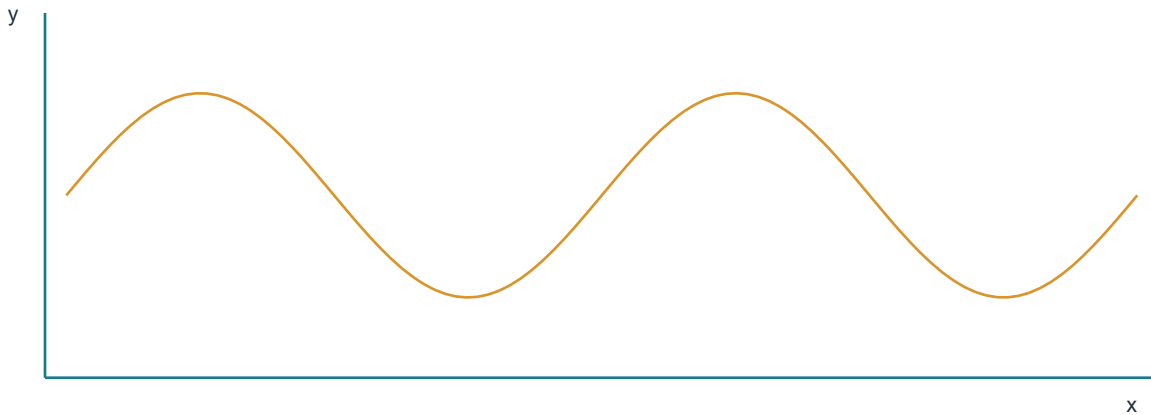
a. Distinguish a pure tone, a periodic complex tone, and noise using a time trace and frequency spectrum. Draw one spectrum for each. b. Explain compression and rarefaction in air. c. If rms pressure doubles, what happens to intensity and sound-pressure level? d. Explain why changing source loudness does not necessarily change sound speed. Also distinguish amplitude, intensity, pitch, loudness, and timbre, and identify which of these a spectrum displays.



8. Waves at boundaries

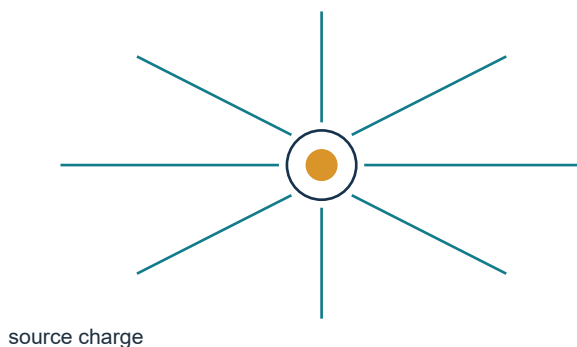
Explain and draw.

- At a boundary, what happens to frequency, speed, wavelength, and direction when a wave enters a slower medium?
- Use Huygens' principle to explain refraction.
- State why the angle is measured from the normal and derive the qualitative direction of bending for $n_2 > n_1$.
- Compare transverse and longitudinal waves, including the role of a medium. Explain why an electromagnetic wave can travel through vacuum while sound cannot, and state which quantities change and which stay continuous at a stationary boundary.



9. Electric-field reasoning with symmetry

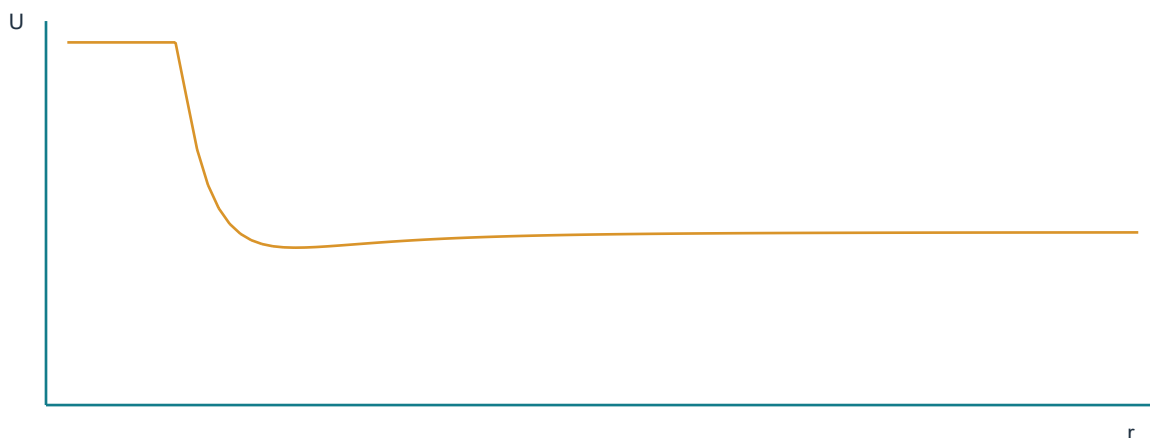
Without calculating a final number, rank the following field magnitudes at the marked point and justify using symmetry, distance, and superposition: (i) one point charge at distance r , (ii) the same charge at $2r$, (iii) two equal like charges symmetrically placed with the point at the midpoint, (iv) two equal opposite charges symmetrically placed with the point at the midpoint. Draw vector contributions for (iii) and (iv). Then state what happens if the observation point is moved slightly off the midpoint, and explain why the inverse-square law alone cannot determine the answer for the two-charge cases.



10. Integrated concept check

Complete each statement and then explain one.

- Electric potential is a _____, while electric field is a _____.
- Moving along an equipotential requires _____ work by the electric field.
- The field points toward _____ potential.
- In a one-dimensional wave, phase difference depends on _____ and _____.
- For a sound wave, pitch is associated mainly with _____, while timbre depends on waveform and _____. f. Capacitance describes _____ per unit _____. g. A dipole in a uniform field can have zero net force but nonzero _____.



11. Comprehensive explanation prompt

Choose two of the following and answer in 200-300 words each, using at least one diagram and one equation per response:

- A. Explain field lines, flux, Gaussian surfaces, and conductor shielding as one connected model.
 - B. Explain how SHM, traveling waves, and sound connect local oscillation to energy transport.
 - C. Explain how a sinusoidal wave's mathematical form encodes amplitude, phase, wavelength, frequency, and direction.
- For every response, add one limiting-case prediction: far from a charge, at a conductor interior, at a turning point, or after entering a slower medium.